GOOGLE CLOUD WEB SERVER DEPLOYMENT ASSIGNMENT

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Course Title:
BIG DATA
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Task 2 – Python Data & File Handling

Objective

The objective of this task is to understand Python data structures, functions, and file handling techniques used in data analysis.

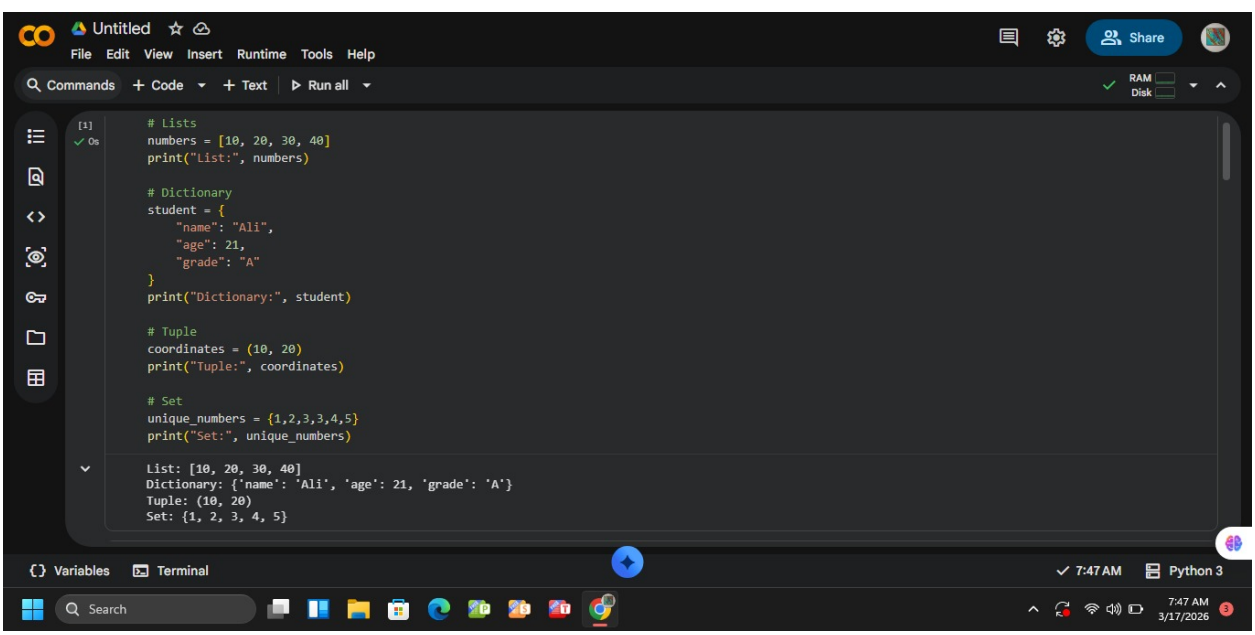
Description

In this task, Python was used to work with different data structures such as lists, dictionaries, tuples, and sets. Functions were created to perform basic data processing operations. File handling techniques were implemented by reading and writing data in CSV and JSON formats. List comprehensions were used for efficient data manipulation, and exception handling was applied to manage runtime errors. A small dataset example was also included to perform simple calculations and demonstrate practical usage.

Outcome

This task helped in developing a strong understanding of Python fundamentals, data handling, and file operations, which are essential for data analysis tasks.

Preview:



[2]

```python
# Function to calculate average marks

def calculate_average(marks):
    total = sum(marks)
    avg = total / len(marks)
    return avg

marks = [70, 80, 90, 85]

average = calculate_average(marks)

print("Average Marks:", average)
```

```
Average Marks: 81.25
```

[3]

```python
import pandas as pd

data = {
    "Name": ["Ali", "Sara", "Ahmed", "Hina"],
    "Math": [78, 85, 90, 88],
    "Physics": [82, 79, 94, 86]
}

df = pd.DataFrame(data)

# Save CSV
df.to_csv("student_grades.csv", index=False)

print("CSV file created")
```

```
CSV file created
```

[4]

```python
df = pd.read_csv("student_grades.csv")

print("Student Data:")
print(df)
```

```
Student Data:
    Name  Math  Physics
0    Ali    78       82
1   Sara    85       79
2  Ahmed    90       94
3   Hina    88       86
```

[5]

```python
import json

student = {
    "name": "Ali",
    "age": 21,
    "course": "Computer Science"
}

# Write JSON
with open("student.json", "w") as file:
    json.dump(student, file)

# Read JSON
with open("student.json", "r") as file:
    data = json.load(file)

print(data)
```

```
{'name': 'Ali', 'age': 21, 'course': 'Computer Science'}
```

[6]

```python
numbers = [1,2,3,4,5]

# Square of numbers
squares = [x**2 for x in numbers]

print("Squares:", squares)
```

```
Squares: [1, 4, 9, 16, 25]
```

[7]

```python
numbers = [10,15,20,25,30]

even_numbers = [x for x in numbers if x % 2 == 0]

print("Even Numbers:", even_numbers)
```

```
Even Numbers: [10, 20, 30]
```

[8]
17s

```python
try:
    num1 = int(input("Enter number: "))
    num2 = int(input("Enter another number: "))

    result = num1 / num2

    print("Result:", result)

except ZeroDivisionError:
    print("Error: Cannot divide by zero")

except ValueError:
    print("Error: Invalid input")

finally:
    print("Program finished")
```

```
Enter number: 4
Enter another number: 3
Result: 1.3333333333333333
Program finished
```

[9]
0s

```python
sales = [1200, 1500, 1100, 1800, 2000]

total_sales = sum(sales)
average_sales = total_sales / len(sales)

print("Total Sales:", total_sales)
print("Average Sales:", average_sales)
```

```
Total Sales: 7600
Average Sales: 1520.0
```